

4

Null constituents

4.1 Overview

So far, our discussion of syntactic structure has tacitly assumed that all constituents in a given structure are **overt** (in the sense that they have overt phonetic features, as well as grammatical and semantic features). However, in this chapter we argue that syntactic structures may also contain **null constituents** (also known as **empty categories**) – i.e. constituents which have grammatical and semantic features but lack phonetic features (and so are ‘silent’ or ‘inaudible’).

4.2 Null subjects

We are already familiar with one kind of null constituent from the discussion of the **Null Subject Parameter** in §1.6. There, we saw that alongside finite clauses like that produced by SPEAKER A in the dialogue in (1) below with an overt subject like *Maria*, Italian also has finite clauses like that produced by SPEAKER B, with a null subject pronoun conventionally designated as *pro* (and referred to affectionately as ‘little *pro*’):

- (1) SPEAKER A: *Maria è tornata?*
 Maria is returned? (‘Has Maria returned?’)
 SPEAKER B: *Sì, pro è tornata*
 Yes, pro is returned (‘Yes, she has returned’)

One reason for positing *pro* in (1B) is that it captures the intuition that the sentence has an ‘understood’ subject (as is clear from the fact that its English translation contains the subject pronoun *she*). A second reason relates to the agreement morphology carried by the auxiliary *è* ‘is’ and the participle *tornata* ‘returned’ in (1). Just as the form of the (third person singular) auxiliary *è* ‘is’ and the (feminine singular) participle *tornata* is determined via agreement with the overt (third person feminine singular) subject *Maria* in (1A), so too the auxiliary and participle agree in exactly the same way with the null *pro* subject in (1B), which (as used here) is third person feminine singular by virtue of referring to *Maria*. If the sentence in (1B) were subjectless, it is not obvious how we would account for the relevant agreement facts. Since all finite clauses in Italian allow a null *pro* subject, we can refer to *pro* as a **null finite subject**.

Although English is not an Italian-style null subject language (in the sense that it is not a language which allows any and every kind of finite clause to have a null *pro* subject), it does have three different types of null subject (briefly discussed in exercise I). One such are **imperative null subjects**. As the examples in (2) below illustrate, an imperative sentence in English can have an overt subject which is either a second person expression like *you*, or a third person expression like *anyone*:

- (2)(a) Don’t *you* dare lose your nerve! (b) Don’t *anyone* dare lose their nerve!

However, imperative null subjects are intrinsically second person, as the contrast in (3) below shows:

- (3)(a) Don’t lose your nerve! (b) *Don’t lose their nerve!

In other words, imperative null subjects seem to be a silent counterpart of *you*. One way of describing this is to say that the pronoun *you* can have a **null spellout** (and thereby have its phonetic features not spelled out – i.e. deleted/omitted) when it is the subject of an imperative sentence.

A second type of null subject found in English are **truncated null subjects**. In cryptic styles of colloquial spoken English (and also in diary styles of written English) a sentence can be **truncated** (i.e. shortened) by giving a subject pronoun like *I/you/he/we/they* a null spellout if it is the first word in a sentence. So, in sentences like those in (4) below:

- (4)(a) *I* can't find my pen (b) *I* think I left it at home (c) Why do I always lose things?

the two italicised occurrences of the subject pronoun *I* can be given a null spellout because in each case *I* is the first word in the sentence, but not other occurrences of *I* – as we see from (5) below:

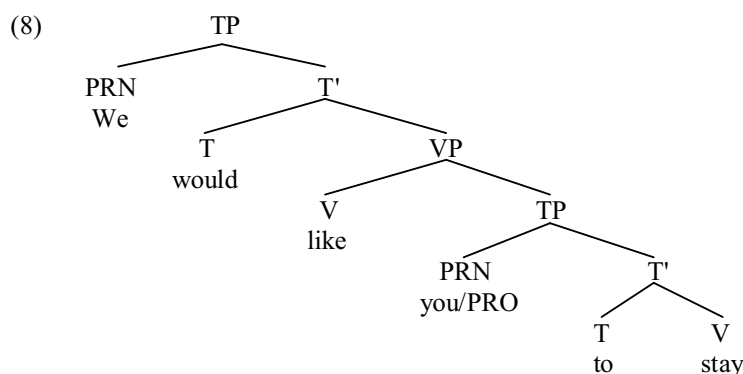
- (5)(a) Can't find my pen (b) Think I left it at home/*Think left it at home
(c) *Why do always lose things?

However, not all sentence-initial subjects can be truncated (e.g. we can't truncate *He* in a sentence like *He is tired*, giving **Is tired*): the precise nature of the constraints on **truncation** are unclear.

A third type of null subject found in English are **nonfinite null subjects**, found in nonfinite clauses which don't have an overt subject. In this connection, compare the structure of the bracketed infinitive clauses in the (a) and (b) examples below:

- (6)(a) We would like [*you* to stay] (7)(a) We don't want [*anyone* to upset them]
(b) We would like [to stay] (b) We don't want [to upset them]

Each of the bracketed infinitive complement clauses in the (a) examples in (6) and (7) contains an overt (italicised) subject. By contrast, the bracketed complement clauses in the (b) examples appear to be subjectless. However, we shall argue that apparently subjectless infinitive clauses contain a **null subject**. The particular kind of null subject found in the bracketed clauses in the (b) examples has the same grammatical and referential properties as a pronoun, and hence appears to be a null pronoun. In order to differentiate it from the null ('little *pro*') subject found in finite clauses in null subject languages like Italian, it is conventionally designated as **PRO** and referred to as 'big PRO'. Given this assumption, a sentence such as (6b) will have a parallel structure to (6a), except that the bracketed TP has an overt pronoun *you* as its subject in (6a), but a null pronoun PRO as its subject in (6b) – as shown below:



Using the relevant technical terminology, we can say that the null PRO subject in (8) is **controlled** by (i.e. refers back to) the subject *we* of the **matrix** (= containing = next highest) clause – or, equivalently, that *we* is the **controller** or **antecedent** of PRO: hence, a structure like 'We would like PRO to stay' has an interpretation akin to that of 'We would like *ourselves* to stay'. Verbs (such as *like*) which allow an infinitive complement with a PRO subject are said to function (in the relevant use) as **control verbs**; likewise, a complement clause with a null PRO subject is known as a **control clause**.

An obvious question to ask at this juncture is why we should posit that apparently subjectless infinitive complements like those bracketed in (6b/7b) above have a null PRO subject. Part of the motivation comes from the intuition that the verb *stay* in (6b) above has an understood subject – and positing a PRO subject for the *stay* clause captures this intuition. The null PRO subject of a control infinitive becomes overt if the infinitive clause is substituted by a finite clause, as we see from the paraphrases for the (a) examples given in the (b) examples below:

- (9)(a) I am sorry [*PRO* to have kept you waiting] (b) I am sorry [*I* have kept you waiting]
(10)(a) Jim promised [*PRO* to come to my party] (b) Jim promised [*he* would come to my party]

The fact that the bracketed clauses in the (b) examples contain an overt (italicised) subject makes it plausible to suppose that the bracketed clauses in the synonymous (a) examples have a null PRO subject.

Further evidence in support of positing a null PRO subject in such clauses comes from the syntax of

reflexive anaphors (i.e. *self/selves* forms such as *myself/yourself/himself/themselves* etc.). As examples such as the following indicate, reflexives generally require a **local antecedent** (the reflexive being italicised and its antecedent bold-printed):

- (11)(a) They want [**John** to help *himself*] (b) ***They** want [John to help *themselves*]

In the case of structures like (11), a local antecedent means ‘an antecedent contained within the same [bracketed] clause/TP as the reflexive’. (11a) is grammatical because it satisfies this locality requirement: the antecedent of the reflexive *himself* is the noun *John*, and *John* is contained within the same (bracketed) *help*-clause as *himself*. By contrast, (11b) is ungrammatical because the reflexive *themselves* does not have a local antecedent (i.e. it does not have an antecedent within the bracketed clause containing it); its antecedent is the pronoun *they*, and *they* is contained within the *want* clause, not within the [bracketed] *help* clause. In the light of the requirement for reflexives to have a local antecedent, consider now how we account for the grammaticality of the following:

- (12) John wants [PRO to prove himself]

Given the requirement for reflexives to have a local antecedent, it follows that the reflexive *himself* must have an antecedent within its own [bracketed] clause. This requirement is satisfied in (12) if we assume that the bracketed complement clause has a PRO subject, and that PRO is the antecedent of *himself*. Since PRO in turn is controlled by *John* (i.e. *John* is the antecedent of PRO), this means that *himself* is **coreferential** to (i.e. refers to the same individual as) *John*.

A further argument in support of positing that control clauses have a silent PRO subject can be formulated in theoretical terms. In the previous chapter, we noted that finite auxiliaries have an [EPP] feature which requires them to have a subject specifier. Since finite auxiliaries belong to the category T of tense-marker, we can generalise this conclusion by positing that all finite T constituents have an [EPP] feature requiring them to have a subject. However, since we argued in chapter 2 that infinitival *to* also belongs to the category T (by virtue of its status as a nonfinite tense-marker), we can suggest the broader generalisation that not only a finite T but also a nonfinite T containing the infinitive particle *to* has an [EPP] feature and hence must likewise project a subject. The analysis in (8) above is consistent with this generalisation, since it posits that *stay* clause either has an overt *you* subject or a null PRO subject, with either type of subject satisfying the [EPP] feature of *to*.

The overall conclusion which our discussion here leads us to is that just as infinitive complements like *you to stay* in (6a) have an overt subject (*you*), so too seemingly subjectless infinitive complements like *to stay* in (6b) have a null PRO subject – as shown in (8) above. In structures like (8), PRO has an explicit **controller**, which is the subject of the matrix clause (i.e. of the clause which immediately contains the control verb). However, this is not always the case, as we can see from structures like (13) below:

- (13)(a) It is important [PRO to take regular exercise]
 (b) It's difficult [PRO to learn a foreign language]
 (c) It's unwise [PRO to mix business with pleasure]

It is clear from examples like (14) below that apparently subjectless clauses like those bracketed in (13/14) must have a null PRO subject:

- (14)(a) It's important [PRO to prepare *myself* properly for the exam]
 (b) It's important [PRO not to take *oneself* too seriously]

since the reflexives *myself/oneself* require a local antecedent within the bracketed clause containing them, and PRO serves the function of being the antecedent of the reflexive. However, PRO itself has no explicit antecedent in structures like (13/14). In such cases (where PRO lacks an explicit controller), PRO can either refer to some individual outside the sentence (e.g. the speaker in 14a) or can have *arbitrary reference* (as in 14b) and refer to ‘any arbitrary person you care to mention’ and hence have much the same interpretation as arbitrary *one* in sentences like ‘*One* can't be too careful these days’. (See Landau 1999/2001 for further discussion of *control* structures.)

4.3 Null auxiliaries

So far, all the clauses we have looked at in this chapter and the last have contained a TP projection headed by a finite auxiliary or infinitival *to*. The obvious generalisation suggested by this is that